GY462

Q1 – Real estate investment is a unique asset class with its specific risks and opportunities.

a. What are the main opportunities of investing in real estate, and how do you 'maximise' these? (10 points)
b. How does your answer change depending on whether you are considering a speculative investment with a horizon of 3-5 years as opposed to a longer-term approach over 10-15 years? (10 points)
c. What are the main risks that characterize to real estate investments, and how do you mitigate these? (10 points)
d. In what ways recent trends on 'green investment' in real estate change the perspective on what the most important risks and opportunities of direct investment in real estate are? (15 points)
e. Indivisibility: what does this term mean in the context of real estate investment? And how does this affect risks and opportunities? (10 points)
f. How does indivisibility affect the construction of a portfolio of real estate assets? And what can an asset manager do in order to mitigate any related issues? (15 points)
g. 'Indirect real estate investment allows to by-pass the issue of indivisibility' – what does it mean? (5 points)
h. Consider REITs: what are the main benefits to investing indirectly in real estate by buying REIT shares? And what are the downsides? (15 points)
i. What are instead the main benefits and costs for asset managers associated with operating as a REIT? (10 points)

Points a- to c- were at the core of the discussions held during Lectures 1 and 2, while point d- was mostly discussed by Dr. Liao in the second half of the term (in her guest intervention) and followed up by two external speakers who presented to GY462 students the perspective from the real estate practice. Points e- and g- were discussed in Lecture 2 and then again during a Lecture on REITs which was jointly delivered by Prof. Silva and Mr. Salway. This lecture also covered points points h- and i-, with Prof. Silva focusing mostly on points e-, g- and h-, while Mr Salway re-iterated some of the key insights required to answer these points and spent a significant part of his intervention discussing point i-. Point f- was discussed by Dr. Szumilo during his lecture on portfolio structuring.

Q2. You are considering buying a house (for owner-occupation) costing £1,500,000, and you are discussing with the bank the terms of a 30-years mortgage. You are a small-business entrepreneur and you estimate your annual income will be £60,000 (net) or £5,000 (net) per month. You also have £750,000 of savings that you could use as a deposit. The bank has notified you that they apply a debt-coverage ratio (DCR) of 1.3 (monthly) and that they would charge an annual interest of 4.5% and apply monthly compounding. The bank also expects to include an origination charge of 1% of the amount you borrow.

a. You are considering a fully-amortizing, constant payment mortgage (CPM) and you would like to use £500,000 of your savings as deposit. What is the monthly cost of this loan? Do you qualify on the basis of the DCR of 1.3? (10 points)

This is a loan for £1mio, over 30 years, with an interest rate of 4.5% and monthly compounding. The cost of this loan is £5,0066.85 so the borrower does not qualify.

b. You decide instead to use all your savings to purchase the property. What is the implied LTV? And what is the monthly payment of the loan? Do you qualify? (10 points)

Using all the savings would bring the borrowed amount down to £750,000 – with an implied LTV of 50%. With 30 years, monthly compounding, and an interest rate of 4.5%, this gives a payment of £3,900.14. A DCR of 1.3 would imply the required income is £4,940 so the borrower qualifies – just about – with an income of £5,000.

c. What are the advantages of using the loan at b- instead of a-? And what are the main downsides/risks? (5 points)

The advantage is that the borrower now qualifies for the loan. The disadvantage is that extra leverage brings in extra risk. Remember: the individual is self-employed so the £5,000 is an expected income. Even small fluctuations might imply a DCR<1.3 – which could alarm the bank. Even larger fluctuations could imply the borrower does not meet the monthly payments and so has to default.

d. What is the effective cost of borrowing of the loan if you hold it to maturity? And what if you repay after 15 years (note: assume you have the fund to repay the outstanding loan)? And what if you repay after 5 years? (15 points)

Remember there is an origination charge of 1%. This means the borrower is actually receiving £750,000 * 99% = £742,500. The effective cost of borrowing if held to maturity is 4.5859%. If the borrower repays after 15 years, the outstanding balance is £496,754.667 and the effective cost of borrowing is 4.6%. If the loan is repaid after 5 years, the outstanding balance is 683,684.38 and the effective cost of borrowing is 4.7344%.

e. What happens to the cost of borrowing when you repay sooner (based on your calculations at point d-)? Comments on your findings. (5 points)

The effective cost of borrowing increases the sooner the loan is repaid. This is because origination charges act as a sunk cost. This point was discussed during Lecture 5.

f. The bank now envisages inflation (previously assumed to be zero) to be 2% per year. What is the interest rate they charge so that they can keep the real value of the payments constant (while the nominal value is eroded by inflation)? [To simplify, determine this annual rate by neglecting monthly compounding] (5 points)

This issue was also discussed in Lecture 5. The right answer is NOT 4.5% + 2%. That is approximate. The right answer can be obtained by using the expression (1.045 * 1.02) – 1 = 6.59%. This is the nominal rate that keeps on an annual basis payment constant (neglecting any monthly compounding).

g. Discuss the 'inflation tilt' problem and clarify in what sense this is non-linear. What problems does it create for borrowers? (5 points)

This problem was discussed in Lecture 5.

h. What is the monthly cost of the new loan at point f- (except for the new interest charges, this loan is the same as in point b- above)? What do you notice? What is the cost of borrowing if you hold it to maturity? (10 points)

The loan with a 6.59% gives a payment of £4,784.99 meaning the borrower does not qualify if the bank is still using a DCR of 1.3. If the loan is held to maturity, the effective cost of borrowing is 6.6874%.

i. In light of these issues, the bank offers an alternative loan. This is structured as an Adjustable Rate Mortgage (ARM) over 30 years and with monthly compounding. The bank agrees to charge an interest rate of 4.5% for the first two-years, climbing to 5.5% at the start of year 3 for another two years; finally stabilizing at 6.5% from the start of year 5 to the end of the loan. What are the three different monthly payments you owe to the bank in the three periods characterized by different interest charges? (15 points)

ARM loans are essentially partitioned CPM. The students first need to work out the monthly payment at 4.5% on the full loan – already obtained above – and the outstanding balance after two years. These are respectively £3,800.14 and 725,245.76. The second tranche is a 28 year mortgage with loan value equal to the outstanding balance and an interest rate of 5.5%. This gives payments of £4,235 and outstanding balance after two years of 702,186. The last calculation requires considering a loan for 26 years with a 6.5% charge applied to this outstanding balance. The payment is £4,668.96.

j. What is the effective cost of borrowing of this loan? Discuss what you notice compared to the loan at point h-. What is the effective cost of borrowing if you repay at the end of year 5? What do you notice? (15 points)

The effective cost of borrowing at maturity can be found using the CF worksheet with the following data. An outflow of money of 742,500 vis a vis 24 months of 3,800.14; 24 months of 4235, and 312 months at 4,668.96. This gives an IRR of 6.0966% which is expectedly lower than the one at point h-. If the ARM is repaid at the end of Year 5, the outstanding balance is 691,485.194. This means that the last part of the CF has 11 months of 4,668.96 and 1 months of 4,668.96 + 691,485.194. This gives an IRR of 5.4676% - which is lower than if the loan was held to maturity. This is slightly counterintuitive compared to was discussed above. But it is explained by the fact that by repaying early the borrower is avoiding long periods of high interest charges.